Graphic Era Hill University
DEHRADUN

IEEE
Graphic Era Hill University
Student Branch

PeerPrep

PLACEMENT READINESS INTELLIGENCE

Practice with purpose.
Prove readiness with evidence.

A unified campus platform for learning, coding, assessments, interviews and actionable skill-gap analytics.

LEARN | ASSESS | IMPROVE

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The Placement Readiness Gap

Preparation is fragmented. Readiness is invisible.

01 Fragmented preparation

Learning content, coding practice, assessments and interviews live in separate tools - so students cannot see one coherent journey.

02 Feedback arrives late

Scores show what happened, but not the next best action. Students repeat activity without a clear, evidence-backed priority.

03 Cohort visibility is weak

Coordinators manually combine attendance, results and feedback, delaying targeted support for students who need it most.

WHO FEELS THE PAIN

STUDENT

What should I improve next?

Scattered feedback -> unclear priorities

COORDINATOR

Who needs help right now?

Manual tracking -> delayed intervention

UNIVERSITY

Is the cohort placement-ready?

Isolated reports -> no shared readiness view

THE OPPORTUNITY

Create one evidence loop: activity -> assessment -> feedback -> targeted preparation -> readiness.

Design principle: decision-ready insights for students and placement teams - not another content library.

One Platform. Three Role-Aware Journeys.

A single workflow turns preparation activity into the next best action.

THE STUDENT LOOP

01 Learn
Structured modules + QuickPrep

02 Practice
Multi-language coding with Judge0

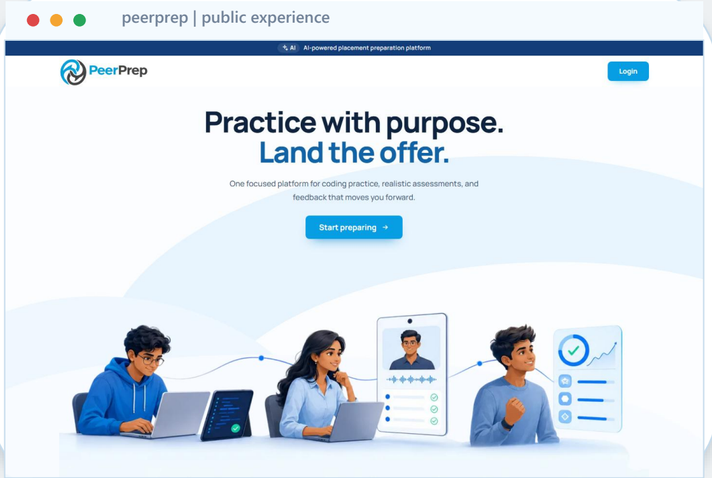
03 Assess
Timed tests + browser monitoring

04 Interview
Peer and AI-assisted practice

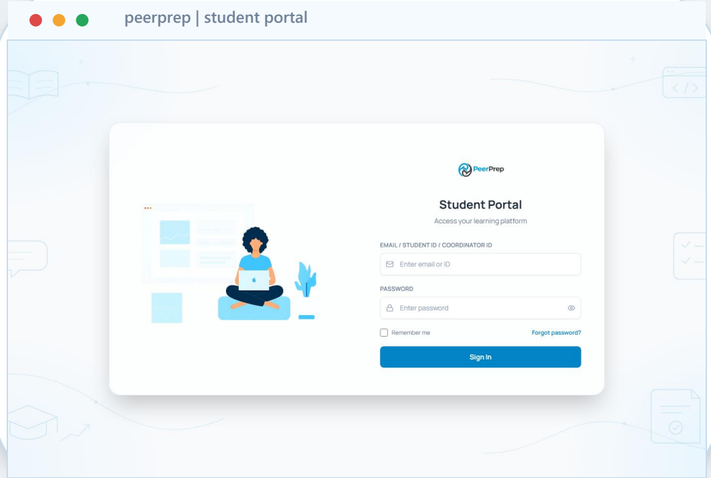
05 Improve
Analytics, feedback and missions

REAL PRODUCT VIEWS

peerprep | public experience



peerprep | student portal



ROLE-AWARE OUTCOMES

STUDENT | NEXT ACTION

COORDINATOR | INTERVENTION

ADMIN | COHORT VIEW

One identity, one progress record, one preparation journey.

From Practice Event to Readiness Insight

Implemented architecture: modular API + independently scalable execution workers.

REQUEST / COMMAND

ACTORS

Role-based users

Students
Coordinators
Administrators

VALIDATE + AUTHORIZE

UI

React 19 + Vite

Dashboards
Coding workspace
Assessment experience

API

Express service

JWT / HttpOnly cookies
RBAC + validation
Rate limits + audit

PERSIST + QUEUE

DOMAIN

Core workflows

Learning | Assessments
Interviews | Analytics
Notifications

DATA

State + events

MongoDB
Valkey / BullMQ
Socket.IO

ASYNC + SPECIALIZED EXECUTION

QUEUE

Tracked job

API creates job ID, applies limits and returns status.

WORKER

Controlled execution

Dedicated worker retries, fails over and applies backpressure.

JUDGE0

Sandboxed code

Compiler verdict, time and memory return to PeerPrep.

INSIGHT

Evidence loop

Results update analytics, notifications and next actions.

RESULT -> PROFILE -> SKILL GAP -> RECOMMENDED NEXT ACTION

Browser-side monitoring uses MediaPipe / TensorFlow.js; authenticated Socket.IO delivers personal notifications.

A Campus-First Model That Scales

Recommended commercialization model

PROPOSED

CUSTOMER	Universities, placement cells and training partners
REVENUE	Annual institutional license + active-student tiers
ADD-ONS	Onboarding, custom assessments and advanced analytics
VALUE	Less manual tracking; earlier, targeted interventions

Why PeerPrep is defensible

01 — Unified evidence graph

Learning, code, assessment, interview and activity signals share one student context.

02 — Operational - not just educational

Admin and coordinator workflows turn insight into assignments, reminders and support.

03 — Safe execution pipeline

Queued jobs, tracked status, limits and Judge0 isolation separate bursty code workloads.

04 — Privacy-aware monitoring

Browser-side proctoring signals reduce the need to stream raw video to generative AI.

CURRENT STACK

React 19 + Vite | Node.js + Express | MongoDB | Valkey / BullMQ | Judge0 | Socket.IO | MediaPipe / TensorFlow.js

Near-term target: a well-bounded modular monolith with independently scalable workers.

Pilot -> Evidence -> Campus Scale

Success metrics for a 90-day pilot

01 Completion

Learning + assessment completion rate

02 Skill gap

Improvement by topic and difficulty

03 Interview

Participation + feedback turnaround

04 Readiness

Change in evidence-backed cohort score

90-DAY EXECUTION ROADMAP

0-30 DAYS

Pilot setup

Select cohort | baseline skills | configure content

31-60 DAYS

Run the loop

Practice | assessments | interviews | interventions

61-90 DAYS

Prove + scale

Compare outcomes | harden ops | campus rollout

THE HACKATHON ASK

Pilot PeerPrep with one placement cohort - and evaluate it on evidence, not promises.

Social: clearer support | Economic: less manual coordination | Environmental: digital-first assessments and reports

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